

Coding & Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub Repository: https://github.com/SatyaSaiGopal/smart-lender Live Hosted App: https://smart-lender-tcjp.onrender.com
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask
Key Features Implemented	- Authentication System: Registration and login functionality (register.html, login.html, login.css). - Data Pipeline: Data analysis and preprocessing (analysis.py, preprocessing.py, preprocessed_data.pkl). - Machine Learning: Model training and prediction inference (train_model.py, loan_model.pkl, scaler.pkl). - Database Management: Backend database operations (database.py). - User Interface & Analytics: Dynamic web pages, result dashboards, and analytics (dashboard.html, analytics.html, predict.html, result.html).
Pending / Incomplete Features	Continuous Integration/Continuous Deployment (CI/CD) pipeline actions (e.g., GitHub Actions) are not currently visible in the repository structure.
Setup / Run Instructions	1. Clone the repository: <code>git clone https://github.com/SatyaSaiGopal/smart-lender</code> 2. Navigate into the directory and create a virtual environment. 3. Install the required dependencies: <code>pip install -r requirements.txt</code> 4. Ensure the dataset/ and model/ directories contain the necessary .csv and .pkl files. 5. Start the web server: <code>python app.py</code> 6. For cloud deployment (like Render), the system utilizes the included Procfile.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

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